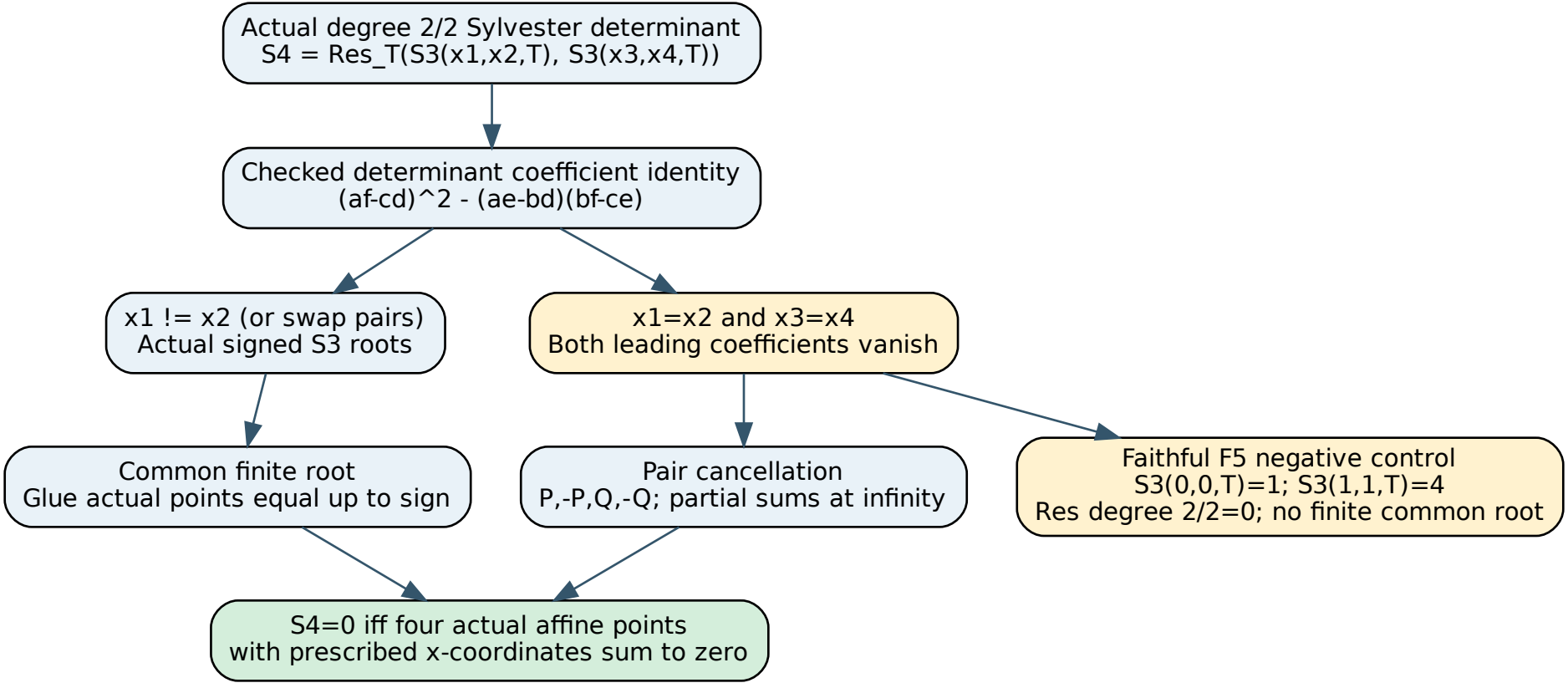
Actual degree 2/2 Sylvester determinant
S4 = Res_T(S3(x1,x2,T), S3(x3,x4,T))
Checked determinant coefficient identity
(af-cd)^2 - (ae-bd)(bf-ce)
x1 != x2 (or swap pairs)
Actual signed S3 roots
x1=x2 and x3=x4
Both leading coefficients vanish
Common finite root
Glue actual points equal up to sign
Pair cancellation
P,-P,Q,-Q; partial sums at infinity
Faithful F5 negative control
S3(0,0,T)=1; S3(1,1,T)=4
Res degree 2/2=0; no finite common root
S4=0 iff four actual affine points
with prescribed x-coordinates sum to zero